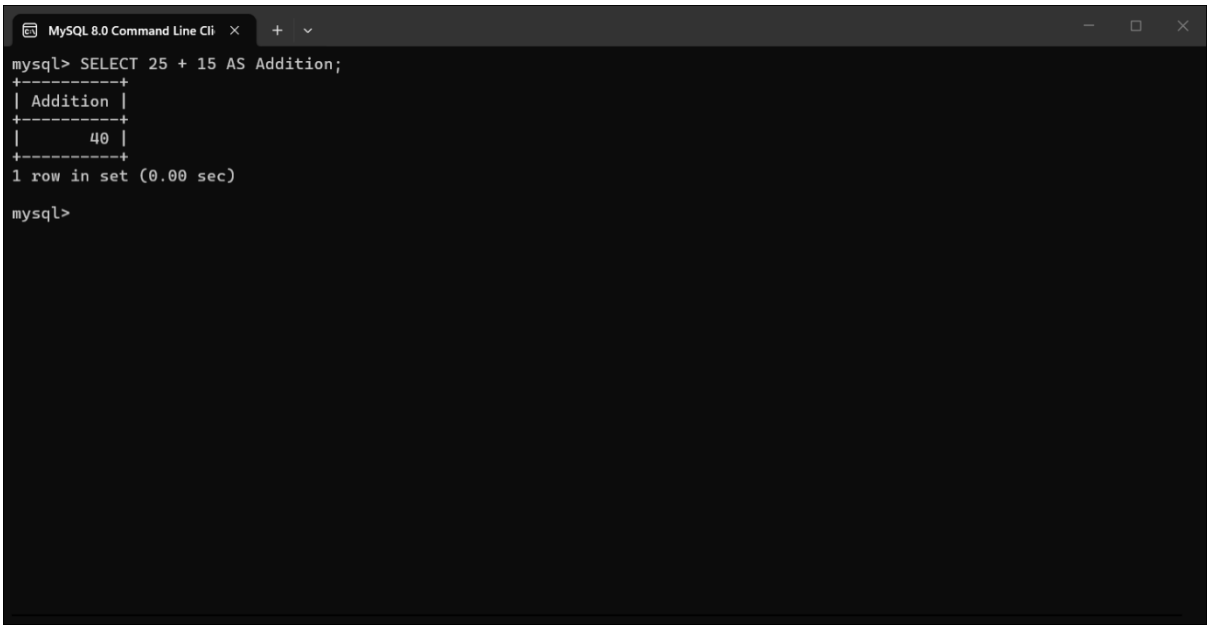
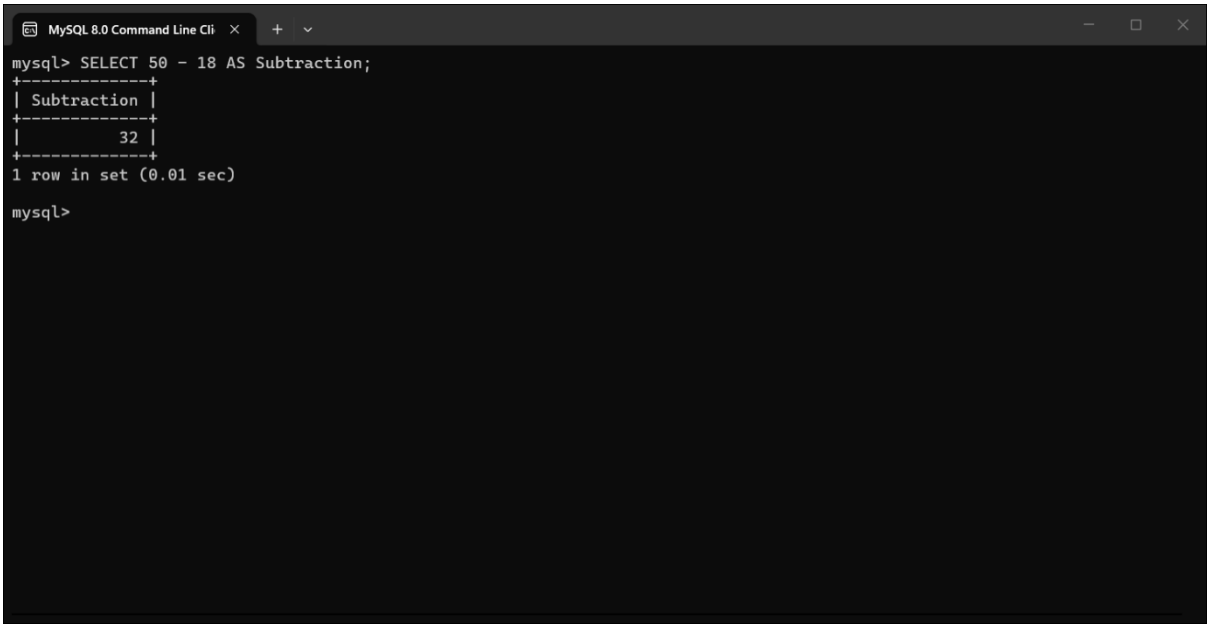
Lab Sheet 4

A. Arithmetic Operators

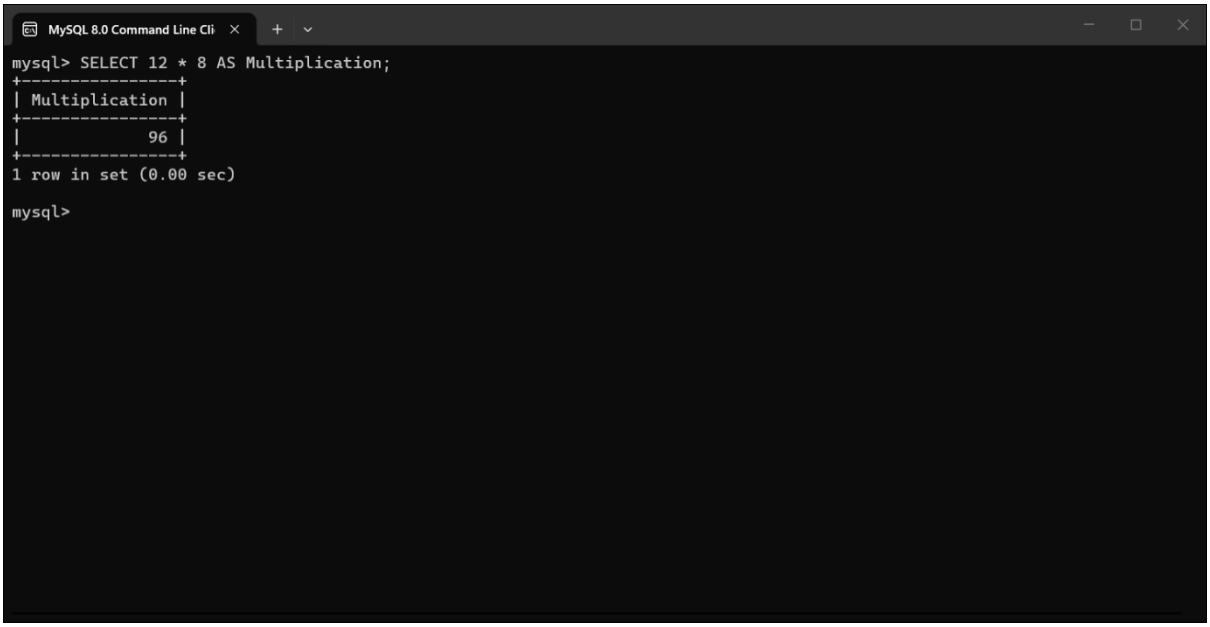
1. Perform addition of two numbers using +.



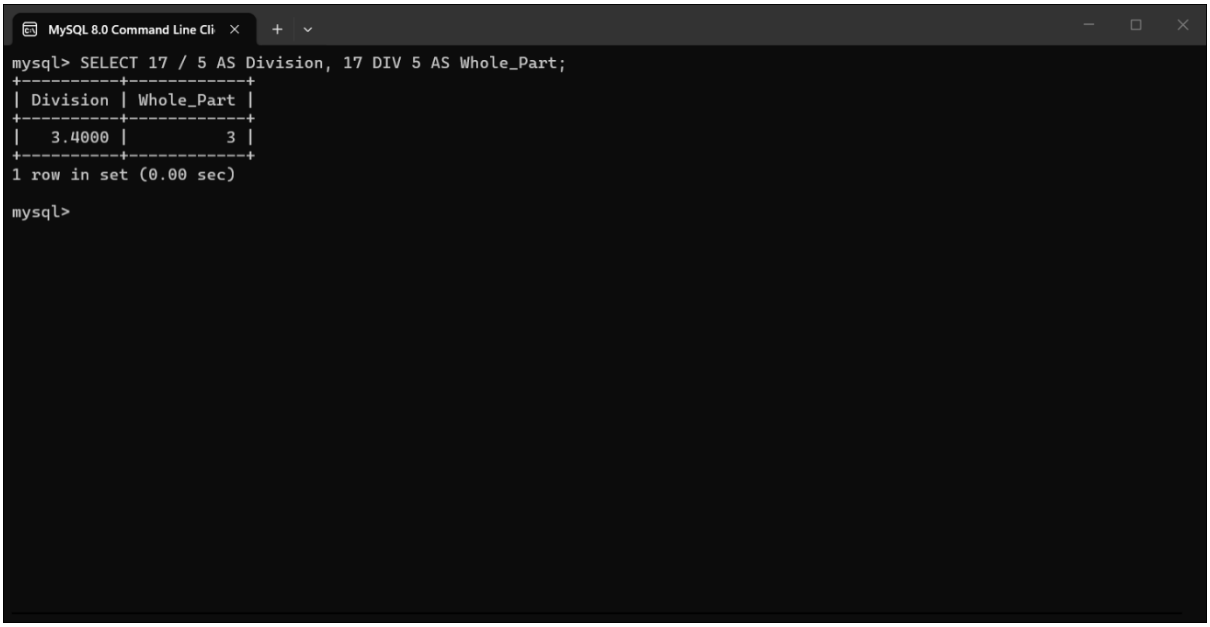
2. Perform subtraction using



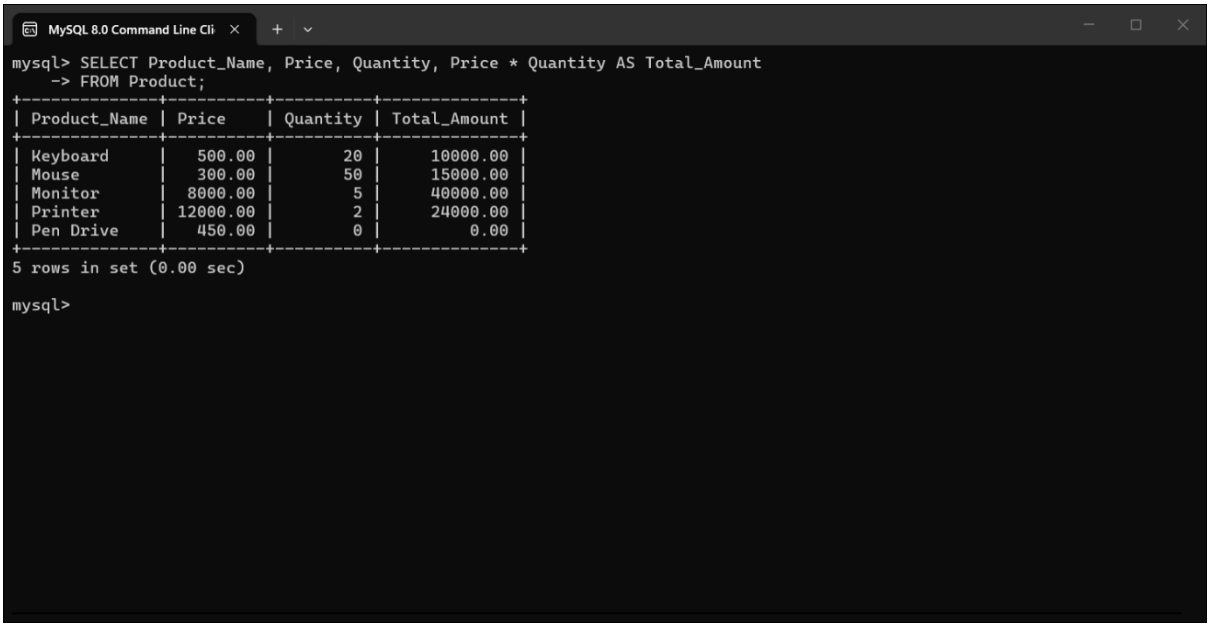
3. Perform multiplication using *



4. Perform division using /.
   Division (/) -- DIV gives only the whole number part

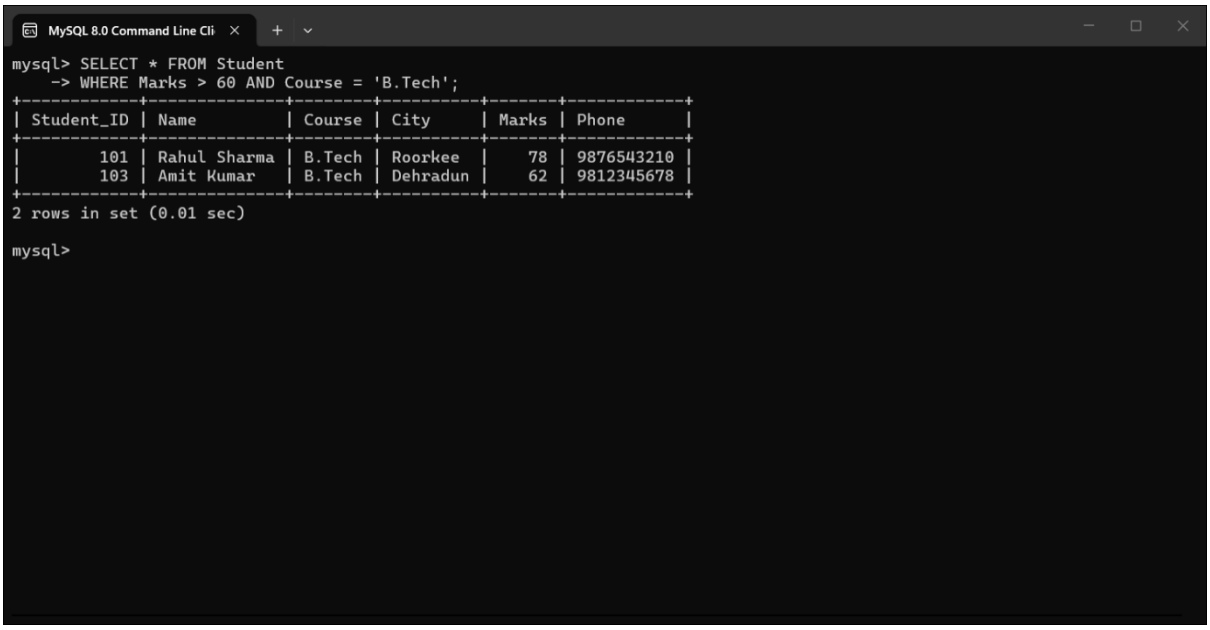


5. Calculate the total amount as Price × Quantity.
   Total amount = Price x Quantity

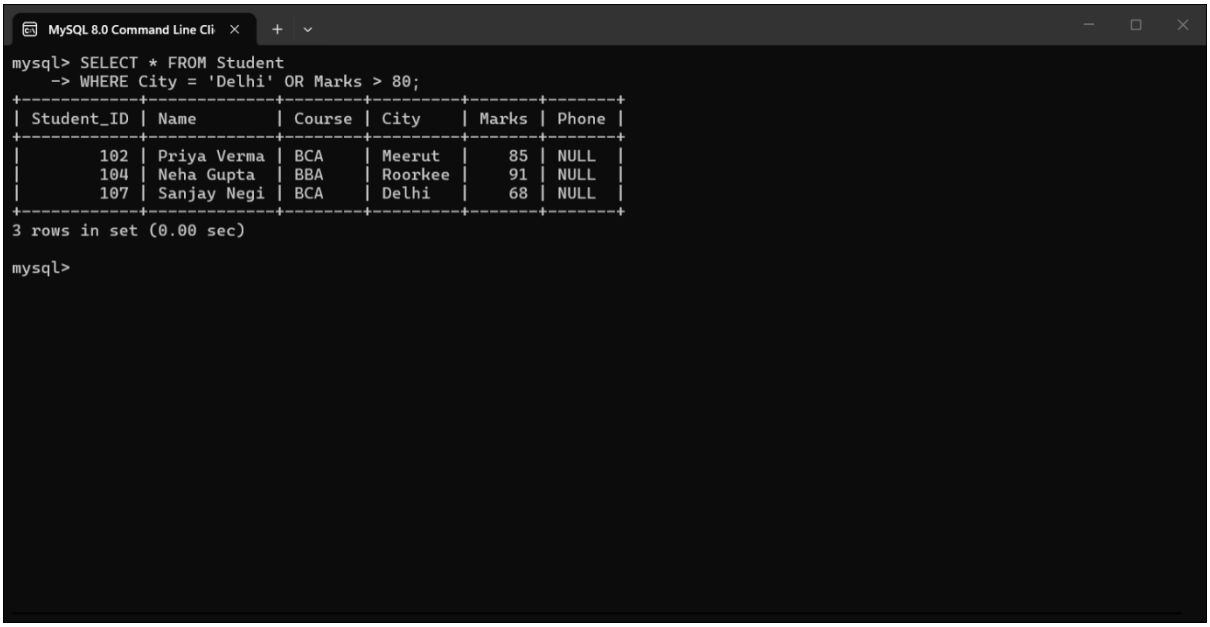


B. Logical operators

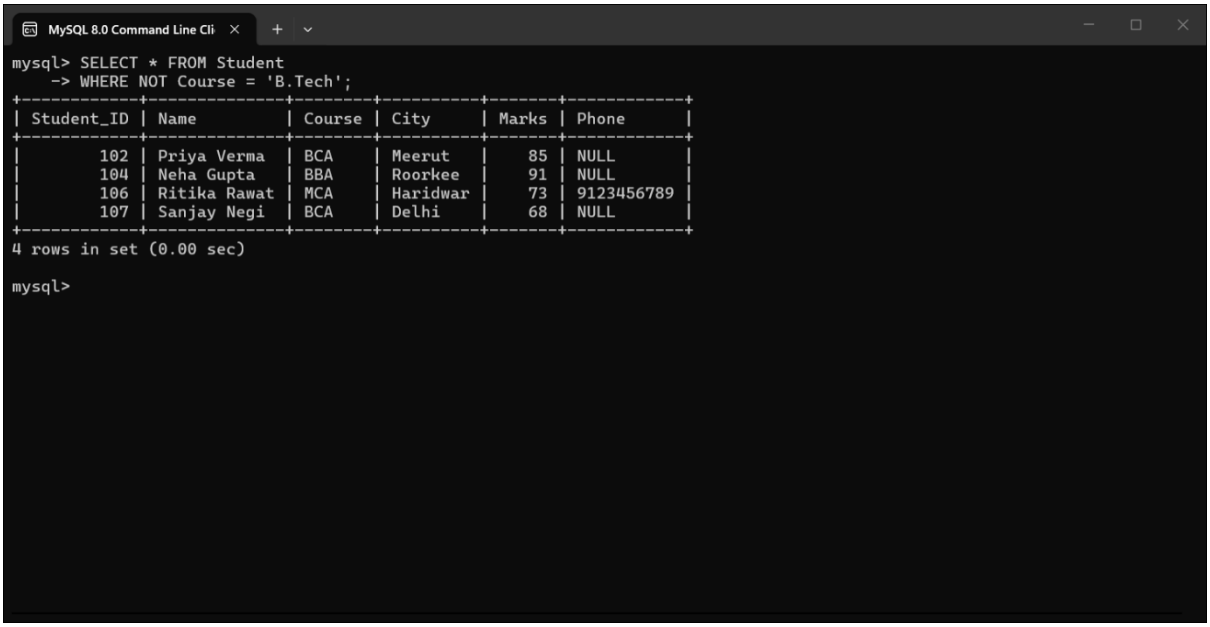
6. Demonstrate AND with two conditions
   AND : both conditions must be true



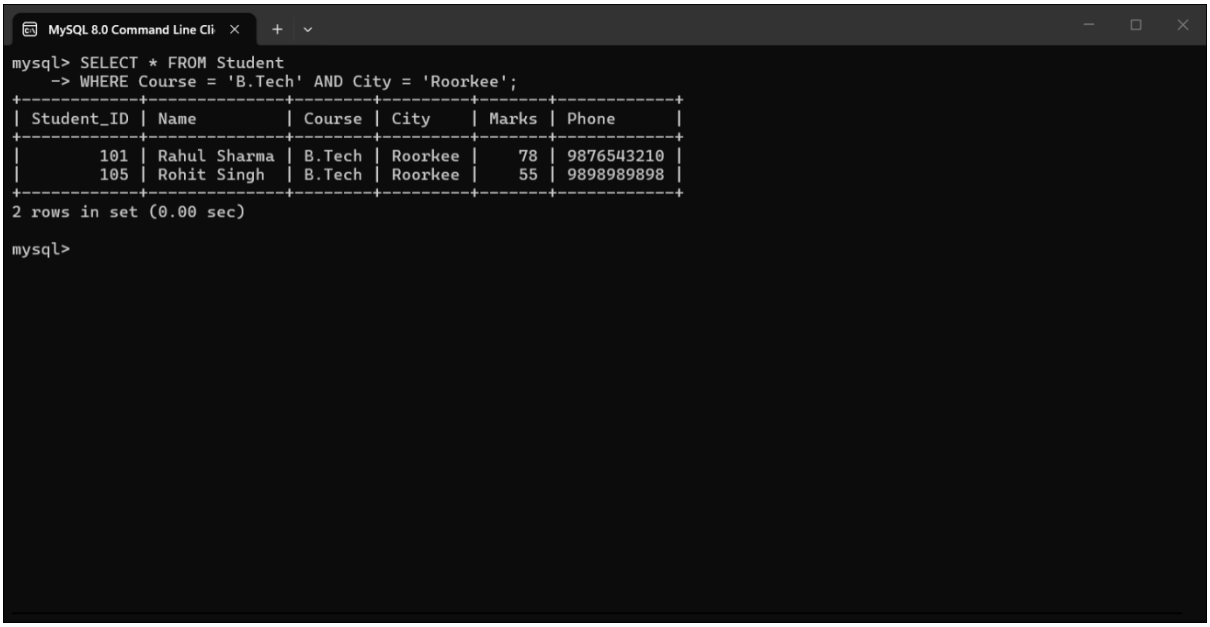
7. Demonstrate OR with two conditions.
   OR : any one condition must be true



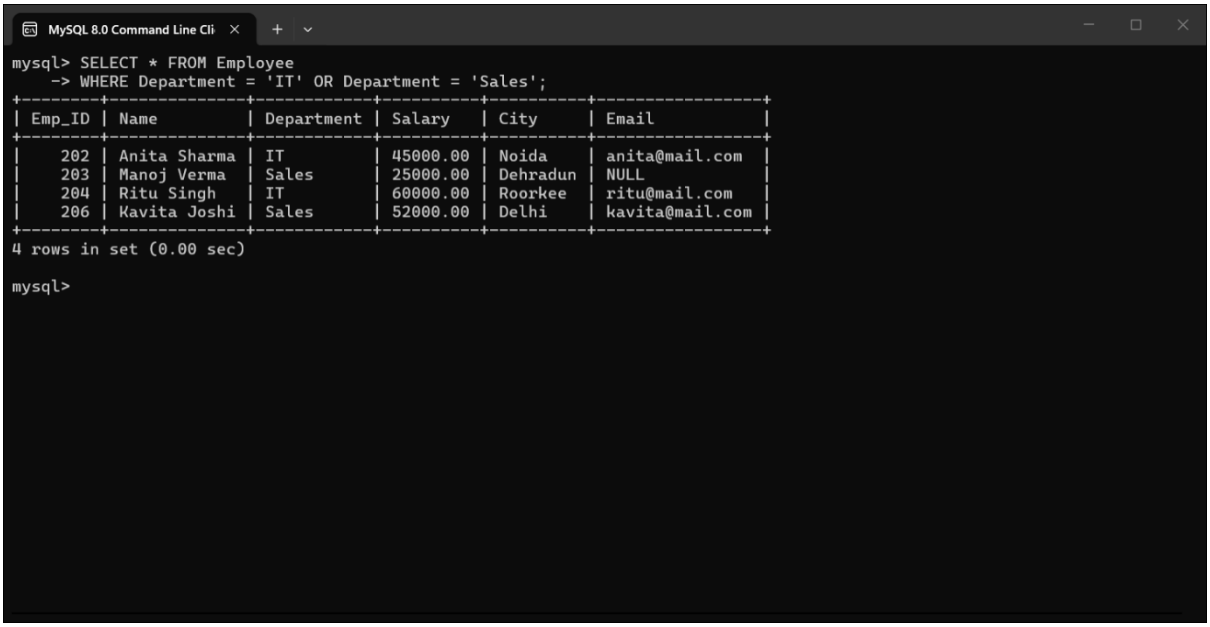
8. Demonstrate NOT with a condition
   NOT : gives rows where the condition is false



9. Find students belonging to a specific course AND city.

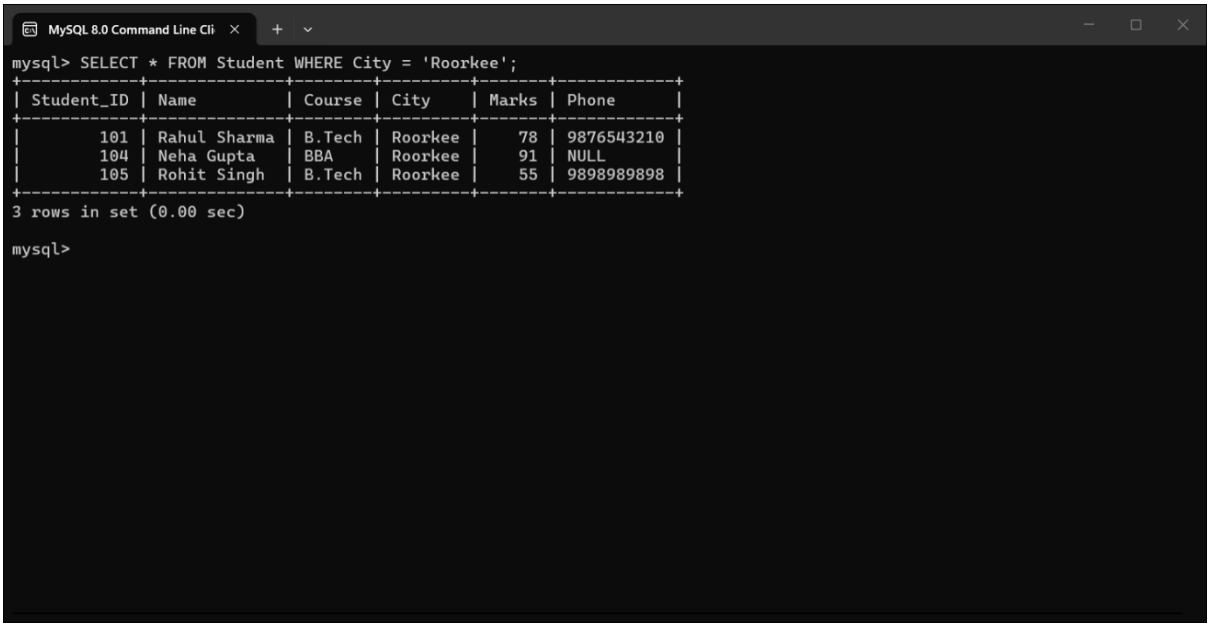


10. Find employees belonging to either of two departments.

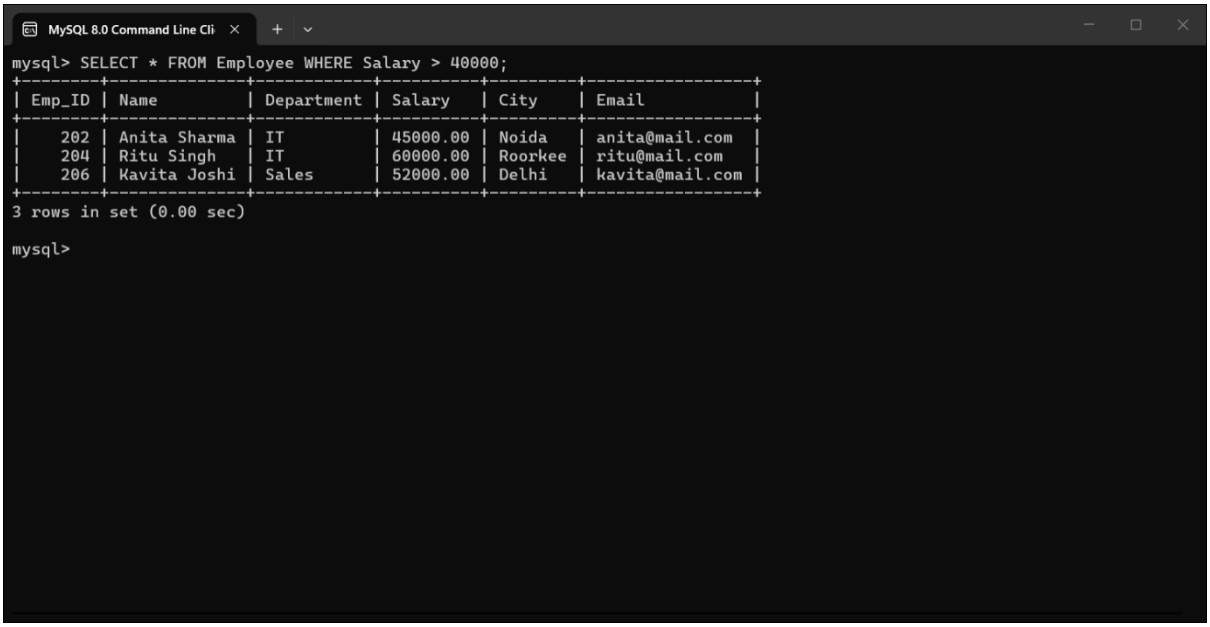


C. Comparison Operators

11. Demonstrate the = operator.



12. Demonstrate the > operator.



13. Demonstrate the < operator.

```
MySQL 8.0 Command Line Cli

mysql> SELECT * FROM Student WHERE Marks < 70;
+------------+-------------+--------+----------+-------+------------+
| Student_ID | Name        | Course | City     | Marks | Phone      |
+------------+-------------+--------+----------+-------+------------+
|        103 | Amit Kumar  | B.Tech | Dehradun |    62 | 9812345678 |
|        105 | Rohit Singh | B.Tech | Roorkee  |    55 | 9898989898 |
|        107 | Sanjay Negi | BCA    | Delhi    |    68 | NULL       |
+------------+-------------+--------+----------+-------+------------+
3 rows in set (0.00 sec)

mysql>
```

14. Demonstrate >= and <=.

```
MySQL 8.0 Command Line Cli

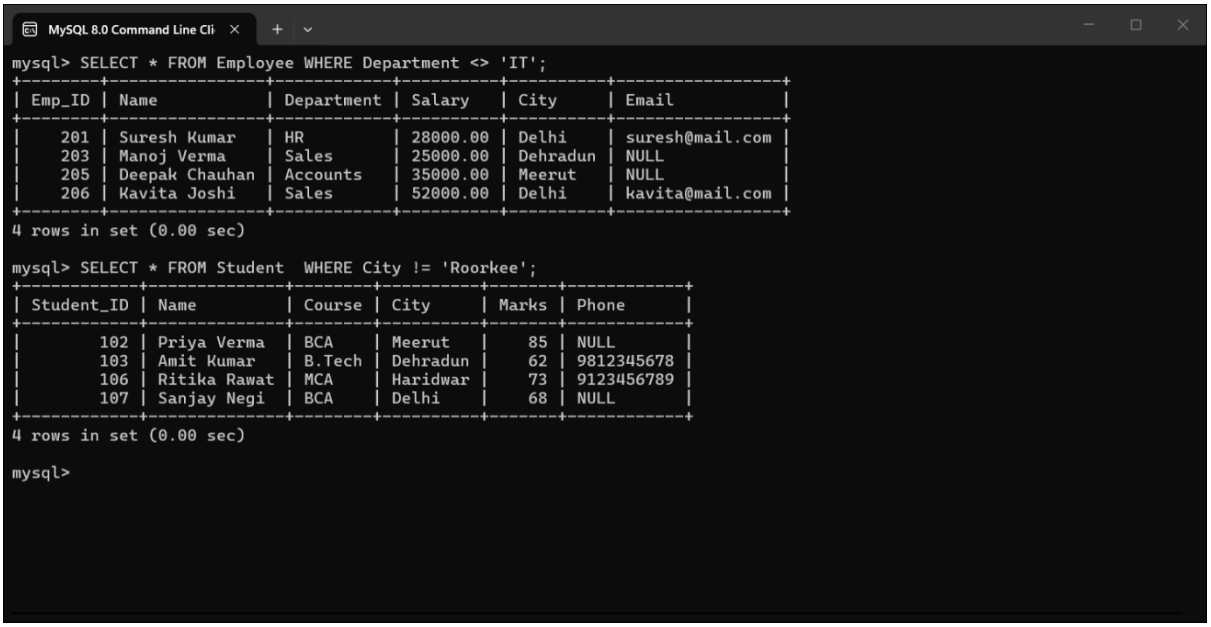
mysql> SELECT * FROM Employee WHERE Salary >= 45000;
+--------+--------------+------------+----------+---------+-----------------+
| Emp_ID | Name         | Department | Salary   | City    | Email           |
+--------+--------------+------------+----------+---------+-----------------+
|    202 | Anita Sharma | IT         | 45000.00 | Noida   | anita@mail.com  |
|    204 | Ritu Singh   | IT         | 60000.00 | Roorkee | ritu@mail.com   |
|    206 | Kavita Joshi | Sales      | 52000.00 | Delhi   | kavita@mail.com |
+--------+--------------+------------+----------+---------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Student  WHERE Marks <= 68;
+------------+-------------+--------+----------+-------+------------+
| Student_ID | Name        | Course | City     | Marks | Phone      |
+------------+-------------+--------+----------+-------+------------+
|        103 | Amit Kumar  | B.Tech | Dehradun |    62 | 9812345678 |
|        105 | Rohit Singh | B.Tech | Roorkee  |    55 | 9898989898 |
|        107 | Sanjay Negi | BCA    | Delhi    |    68 | NULL       |
+------------+-------------+--------+----------+-------+------------+
3 rows in set (0.00 sec)

mysql>
```

15. Demonstrate the <>/!= operator.

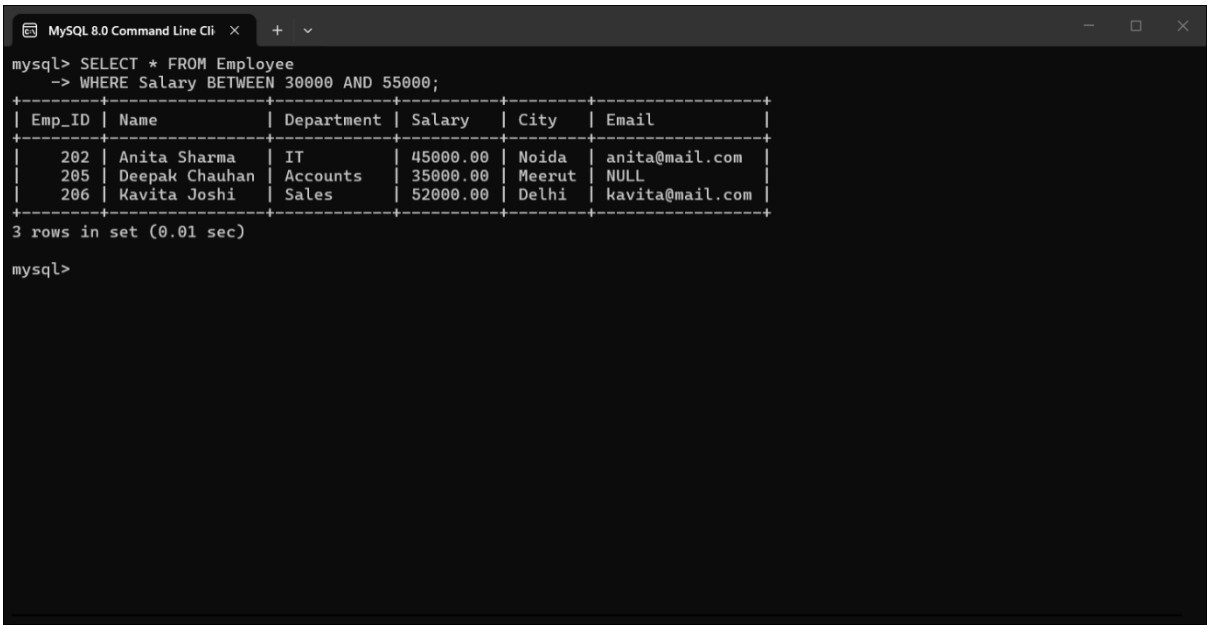
    <> and != (not equal to) : both work the same



D. Special Operators

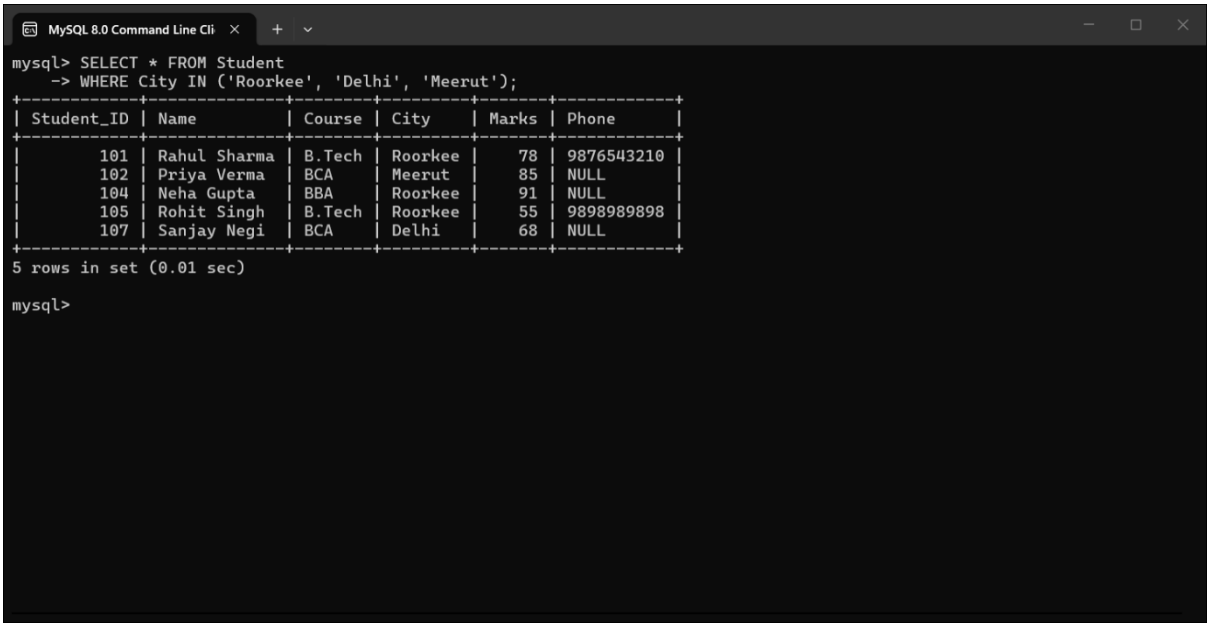
16. Demonstrate BETWEEN for salary range.

    BETWEEN : salary in a range (both ends included)



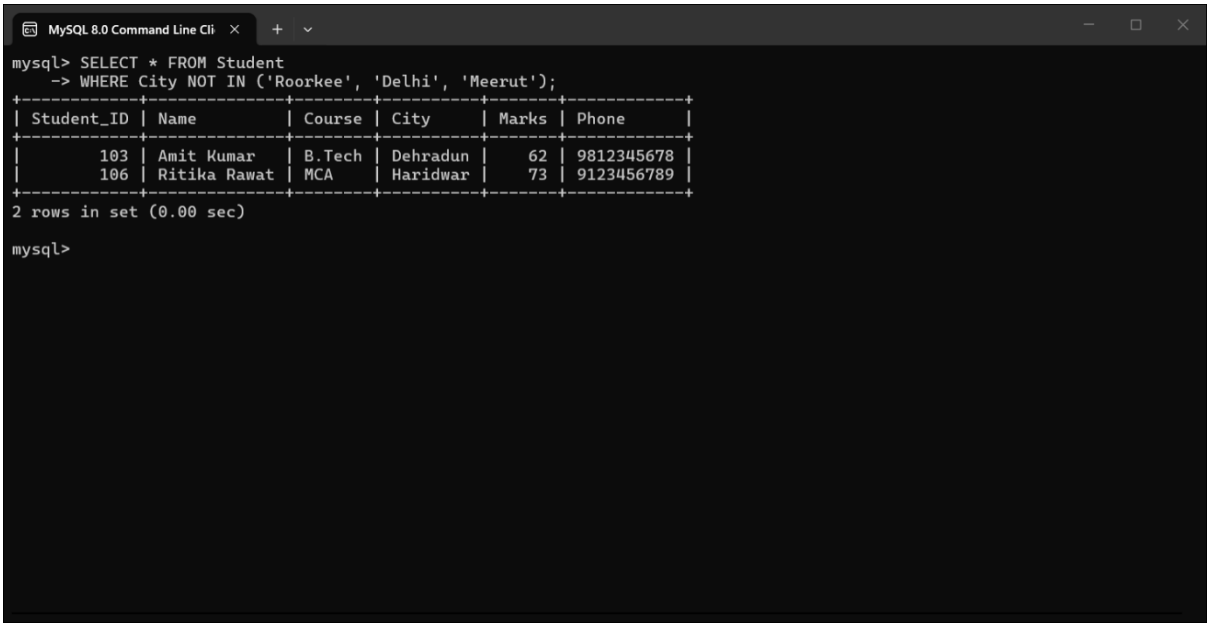
17. Demonstrate IN for selecting multiple cities.

    IN : students from a list of cities



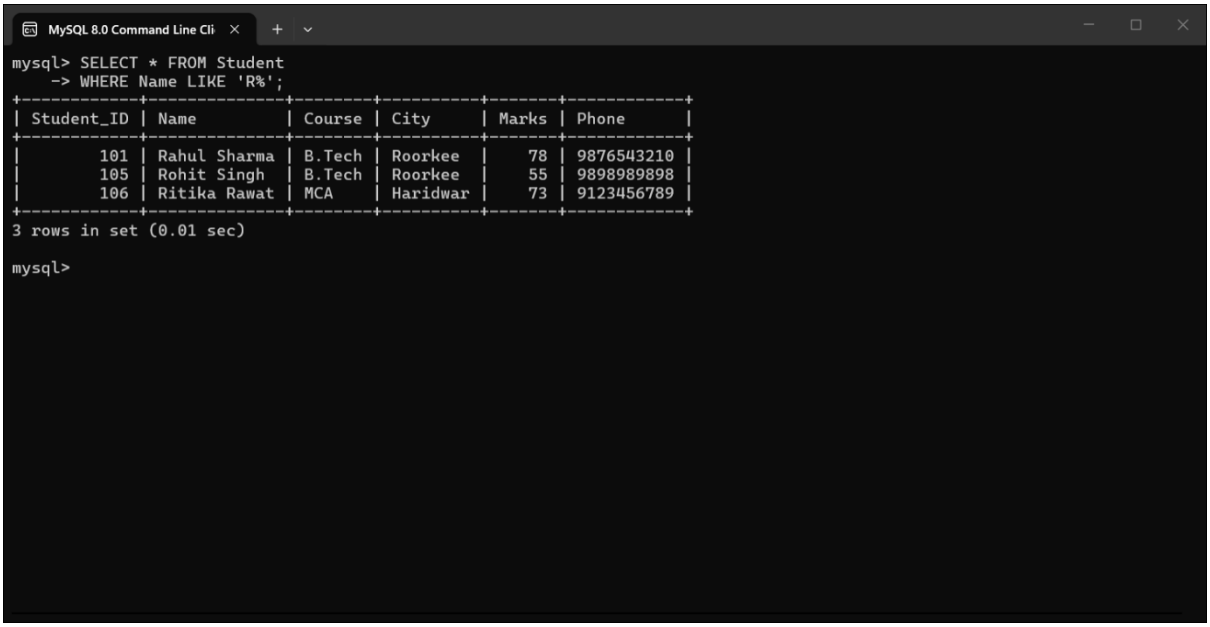
18. Demonstrate NOT IN.

    NOT IN : students not from these cities

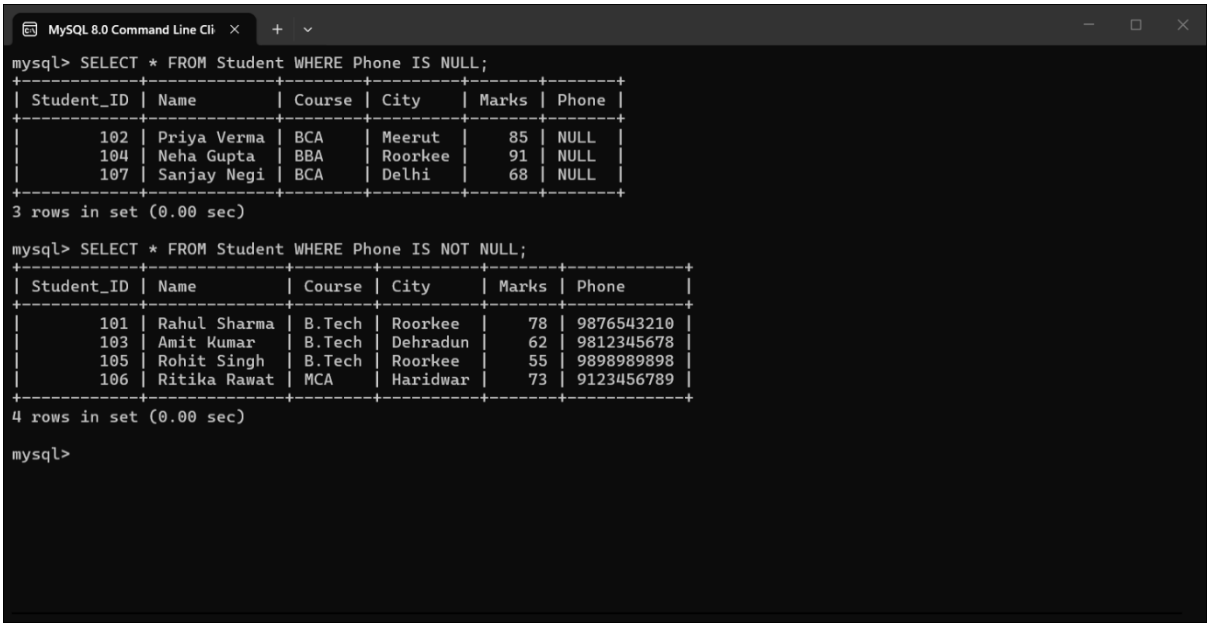


19. Demonstrate LIKE for names beginning with a particular letter.

    LIKE : names starting with the letter R% means any number of characters



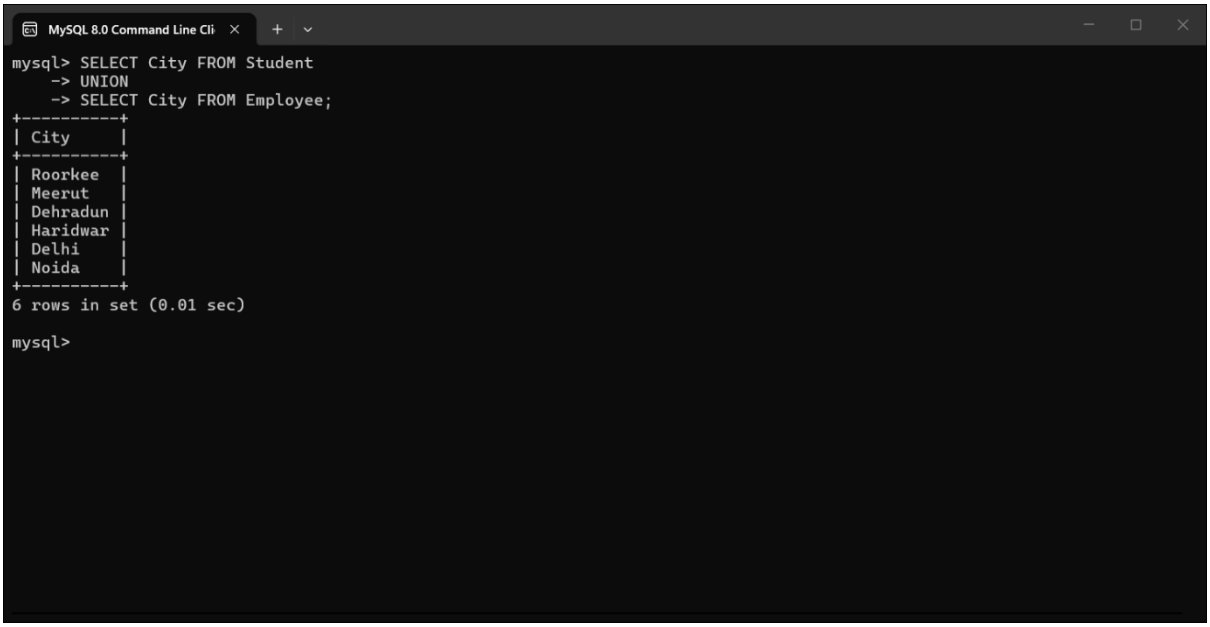
20. Demonstrate IS NULL and IS NOT NULL.



E. Set Operators

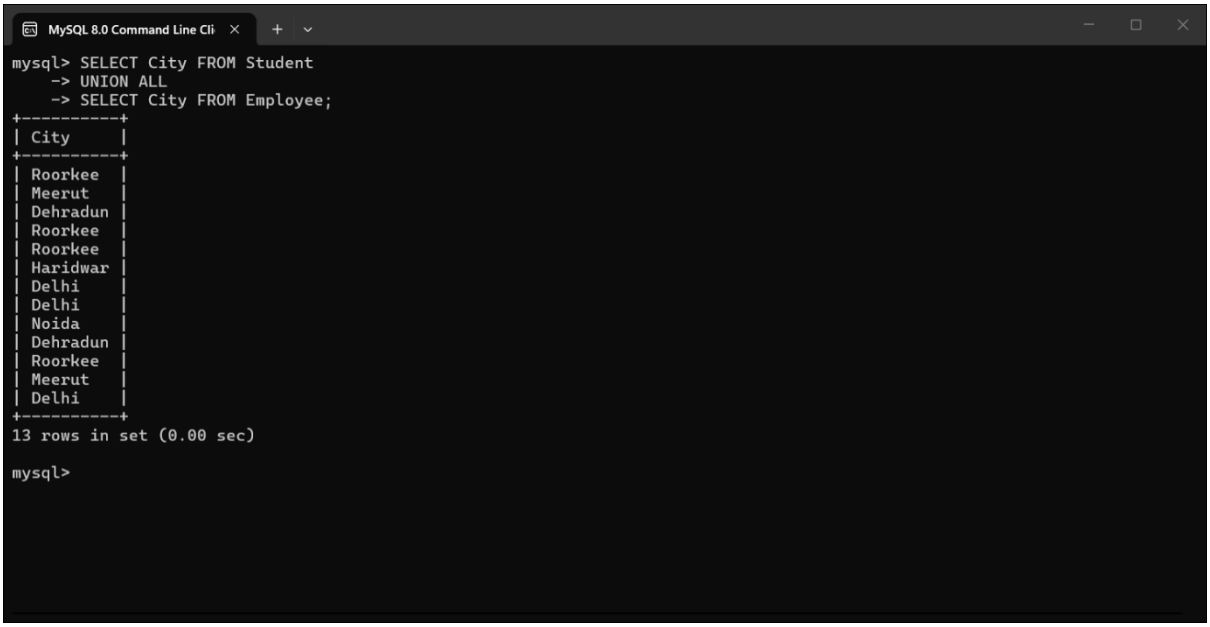
21. Demonstrate UNION using two compatible queries.

UNION : joins both results and removes repeated values



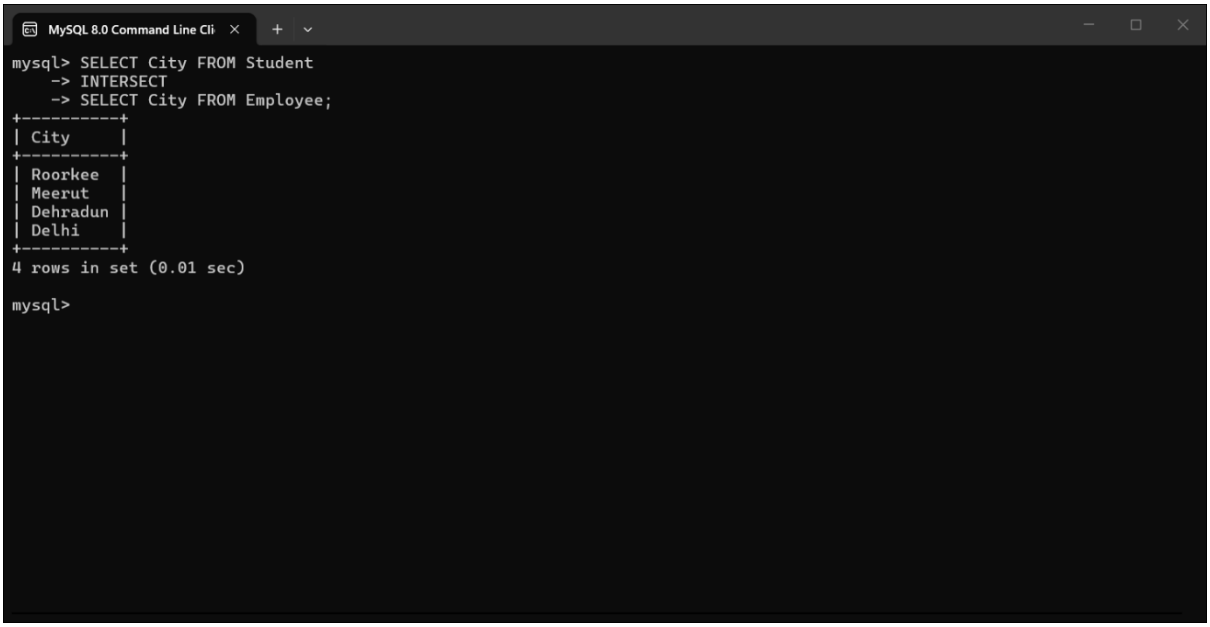
22. Demonstrate UNION ALL.

    UNION ALL : joins both results and keeps repeated values

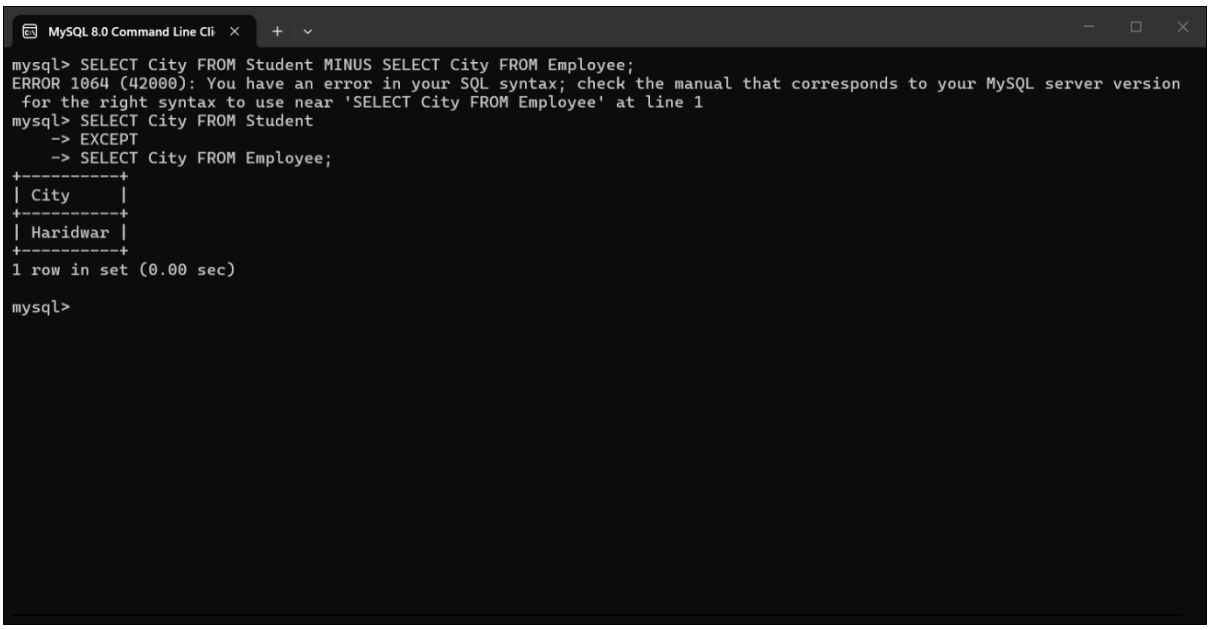


23. Demonstrate INTERSECT where supported by the DBMS

    INTERSECT : cities present in both tables (works in MySQL 8.0.31 and above)



24. Demonstrate EXCEPT/MINUS where supported.
    EXCEPT : cities of students that are not in Employee table (works in MySQL 8.0.31 and above)
    MySQL does not have MINUS. In Oracle the same query is written with MINUS:



25. Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operators.

    Using many operators together
    Find IT or Sales employees whose yearly salary is more than 4,00,000, who have an email, and who are not from Dehradun.

-- Arithmetic : Salary * 12

-- Logical : AND

-- Comparison : > and <>

-- Special : IN and IS NOT NULL

